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## **Explaining Deep Learning Models in Full Waveform Inversion: Enhancing Transparency in Seismic Data Interpretation**

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### **Abstract**

We present a physics-informed neural network approach for full-waveform inversion (FWI) that embeds a Deep Image Prior (DIP) U-Net within the inversion loop and applies explainable AI (XAI) methods to interpret the learned model. Starting from a decimated Marmousi velocity model, we generate synthetic seismic data via 2D acoustic wave propagation. A smoothed version of the true model initializes a randomly initialized U-Net (the DIP), which is pretrained to replicate the smoothed model. Thereafter, the DIP's weights replace the cell-wise velocity parameters in FWI, and an iterative optimizer adjusts these weights to minimize discrepancies between observed and simulated waveforms. The resulting high-resolution velocity model arises from physics-constrained learning. To reveal the network's decision process, we compute attribution maps using SHapley Additive exPlanations (SHAP), Integrated Gradients, and occlusion sensitivity. These maps highlight the spatial features; primarily high-contrast layers and anomalies that drive the DIP's output. Comparative analyses of true vs. predicted models, residuals, and combined depth/lateral sensitivity profiles confirm that the deep model captures essential subsurface structures. This framework enhances transparency in seismic inversion by linking data-driven models to geophysical knowledge, thus bridging the gap between "black-box" deep learning and interpretable imaging.

**Keywords:** Full Waveform Inversion, Deep Learning, Explainable AI, Seismic Imaging, Model Interpretability

### **Introduction**

Full-waveform inversion (FWI) represents a nonlinear optimization technique that progressively refines a subsurface velocity model to ensure that the simulated seismic wavefields align with the observed data (Zerafa et. al., 2025). In its acoustic variant, the forward problem is dictated by the two-dimensional acoustic wave equation:

$$\frac{1}{v(x, z)^2} \frac{\partial^2 p(x, z, t)}{\partial t^2} - \nabla^2 p(x, z, t) = s(x, z, t) \quad \text{eq.1}$$

where  $p$  is pressure,  $v(x,z)$  is the velocity field, and  $s$  is the source. Given observed data  $d_{obs}$  on receivers ( $r$ ) and a trial model ( $m$ ) encoding  $v$ , FWI minimizes a misfit functional, typically the  $L_2$  norm;

$$J(m) = \frac{1}{2} \sum_{r,t} (p_{pred}(m) - p_{obs})^2 \quad \text{eq.2}$$

across all shots and receivers. Gradient-based updates (such as adjoint-state gradients) subsequently refine  $m$  until the data residuals are minimized (Virieux and Operto, 2009). Conventional FWI may experience issues related to nonlinearity and insufficient resolution, which has led to a growing interest in enhancing it with learned models (Zerafa et. al., 2025).

Deep learning models, such as convolutional neural networks, have found applications in geophysical inversion; however, their inherent black-box characteristics impede interpretability. Explainable AI (XAI) methodologies strive to "uncover" these black boxes by linking predictions to their corresponding inputs (Lundberg and Lee, 2017; Zerafa et. al., 2025). Specifically, SHAP (SHapley Additive exPlanations) determines feature significance through the lens of cooperative game theory (Lundberg and Lee, 2017). Integrated Gradients calculates the path integral of gradients extending from a baseline to the input (Zerafa et. al., 2025). Occlusion sensitivity methodically alters segments of the input to observe variations in the output. While these techniques have been employed to interpret image classifiers and various deep networks, their application in physics-driven inversions represents a novel approach. In this study, we integrate these concepts. We employ a Deep Image Prior (DIP) within an Automatic Differentiation-Based Full Waveform Inversion (ADFWI) framework presented by Liu et al., (2025), as the underlying network. Ulyanov et al. (2018) demonstrated that a CNN initialized randomly can serve as a prior for inverse problems without the need for training. We train the DIP to conform to a smooth velocity model, subsequently utilizing the network within Full Waveform Inversion (FWI). Given that the DIP is deep yet untrained on external datasets, it remains aligned with the physical constraints inherent to FWI. We then implement SHAP, Integrated Gradients, and Occlusion techniques on the DIP to determine which spatial features in the velocity model have the most significant impact on the predictions. This process produces visual 'attribution maps' that elucidate the inversion results in terms of physical significance.

## Methodology

We construct the workflow in four distinct phases: model setup, synthetic data generation, DIP-U-Net training, and DIP-FWI inversion.

### Velocity Model

Our starting point is the 2D Marmousi benchmark model (Liu et al, 2025), which has been decimated and transformed into compressional velocity ( $v_p$ ) and density ( $\rho$ ). The high-frequency Marmousi model serves as a complex and realistic test case. We downsample the model by a factor of 20 to ensure that computations remain manageable. The resulting true model  $v_{true}(x,z)$  (Figure 1a) has dimensions of  $n_z \times n_x$  ( $36 \times 137$  grid) and includes layers and salt-like structures. From  $v_{true}$ , we derive an initial smooth model  $v_{smooth}$  by applying a Gaussian filter, which is a common practice in FWI initialization (Liu et al, 2025). This  $v_{smooth}$  serves as our initial point for the DIP pretraining.

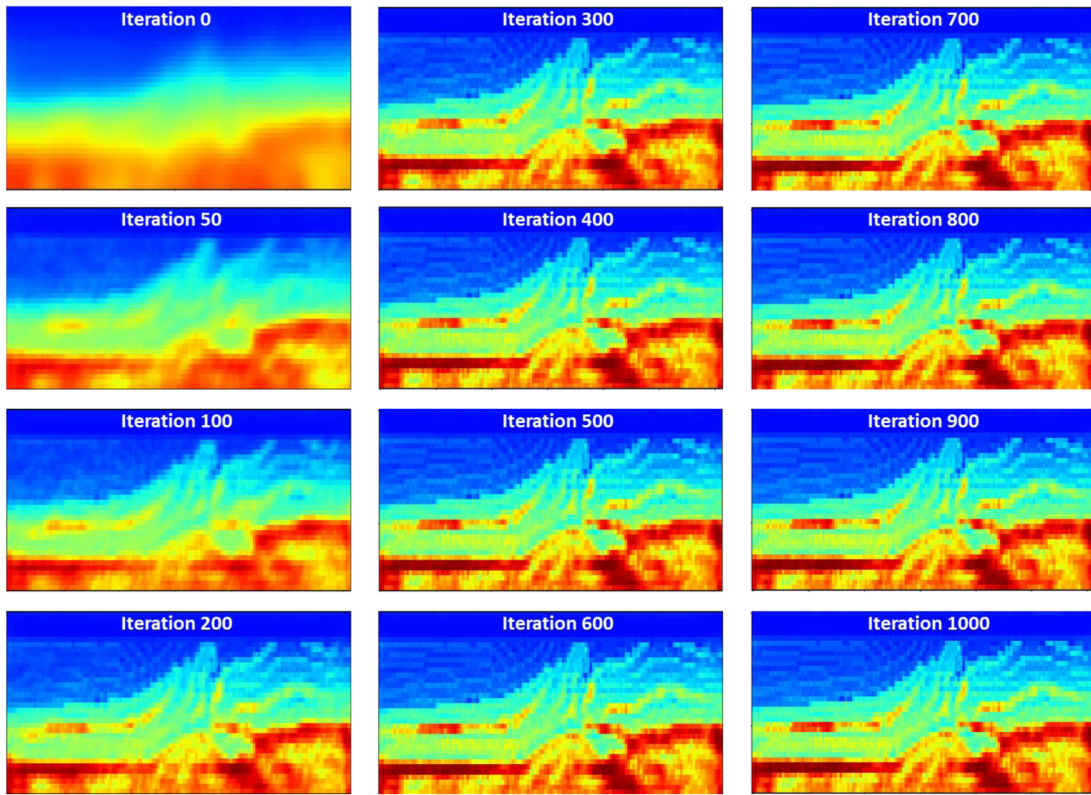


Figure 1—Predicted models at selected iteration intervals during the first 1000 inversion steps.

### Synthetic Data

We establish a survey consisting of vertical (exploding reflector) sources positioned close to the surface, along with a line of receivers located at the top. The acoustic wave equation is solved numerically (using ADFWI propagator framework) within this velocity model (Liu 2024; Liu et al, 2025). A Ricker wavelet ( $f_0 = 5 \text{ Hz}$ ) is utilized as the source time function. For each source, we record simulated pressure wavefields  $p_{pred}(x, z, t)$  along the receiver line to produce the observed data  $d_{obs}$ . This data is then used as the "observed" data for the inversion process.

### DIP-UNet Pretraining

We utilize a UNet architecture, proposed by Liu 2024, as the DIP generator, where random tensor noise  $z$  is fed into the network, which subsequently outputs a velocity image  $v_\theta(x, z)$ , parameterized by weights  $\theta$ . Initially,  $\theta$  is set to random values, resulting in an unstructured  $v_\theta$ . We pretrain  $\theta$  by minimizing the loss function;

$$L_{pre}(\theta) = \|v_\theta(x, z) - v_{smooth}(x, z)\|^2 \quad \text{eq.3}$$

using Adam optimization over 20,000 iterations. This process enables the DIP to capture the large-scale structure of  $v_{smooth}$ . Essentially, the network learns the smooth subsurface trend, functioning as a learned prior (Liu 2024; Liu et al, 2025). By the end of pretraining, we find that  $v_\theta$  is approximately equal to  $v_{smooth}$ .

### FWI with DIP

We subsequently establish the network architecture and integrate it into ADFWI framework (Liu 2024; Liu et al, 2025). Rather than modifying the velocity values at each grid cell, we focus on optimizing the network weights  $\theta$ . The model is now represented as  $v_\theta(x, z)$ , which is produced by the DIP. We create a new model object where  $v$  is determined by the output of the DIP, and the density  $\rho$  may either be jointly

estimated or kept constant. The physics solver (Acoustic Propagator within the ADFWI framework) utilizes  $v_\theta$  to simulate wavefields (Liu 2024; Virieux and Operto, 2009). The FWI loss function is defined as

$$J(\theta) = \frac{1}{2} \sum_{shots} \|d_{obs} - d_{pred}(\theta)\|^2 \quad \text{eq.4}$$

where  $d_{pred}(\theta)$  represents the waveforms simulated using  $v_\theta$ . We calculate the gradients of  $J$  with respect to  $\theta$  through automatic differentiation applied to both the solver and the network (Liu 2024; Liu et al, 2025). An Adam optimizer, equipped with learning-rate scheduling, operates for over 1000 iterations. Additionally, we implement a gradient mask to nullify updates in water layers (if present) to maintain the known p-wave velocity of water. Every 50–100 iterations, we preserve intermediate models  $v_\theta$  (Figure 1). The inversion process reaches convergence when the waveform misfit stabilizes, resulting in a final estimate of  $v_{pred}(x,z) = v_{\theta^*}(x,z)$ . Figure 2b illustrates the final predicted model. The residuals  $v_{pred} - v_{true}$  are calculated for further analysis (Figure 2c).

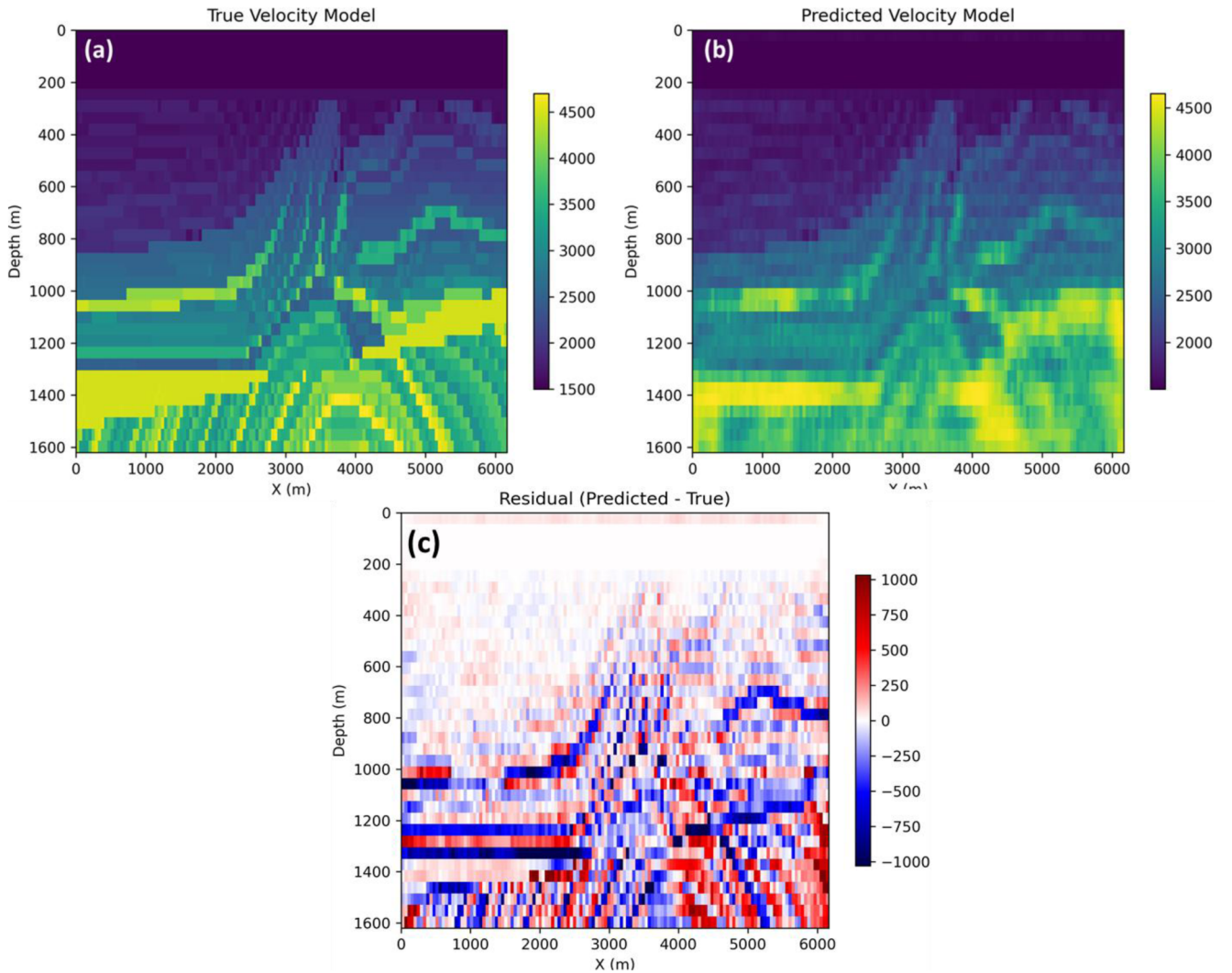


Figure 2—(a) The actual compressional-wave velocity model  $v_{true}(x,z)$ . (b) The model predicted by DIP-FWI  $v_{pred}(x,z)$ . (c) The residual calculated as  $v_{pred} - v_{true}$ . The main structures are accurately retrieved, and the residuals are minimal.

## Explainability Analysis

Following the inversion process, we implement XAI techniques on the DIP model to explain its output. The network translates input latent  $z$  into output velocity  $v_\theta(x,z)$ . To utilize SHAP and IG, which necessitate



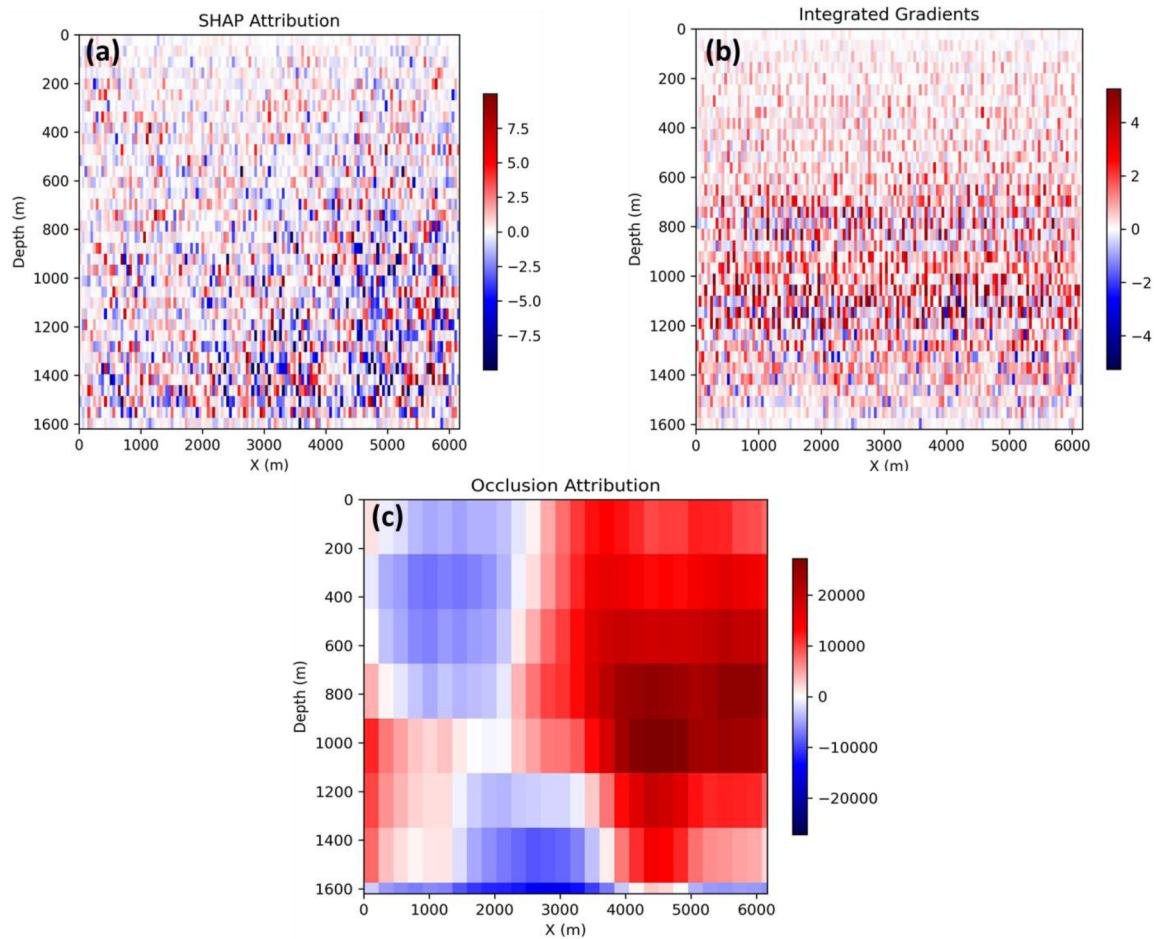
a scalar output, we modify the U-Net to produce the spatial average of velocity, resulting in a single value derived from all contributing pixels. We create several random latent inputs, referred to as ‘background’, alongside one test input, which consists of the base latent input augmented with minor noise, in accordance with established protocols (Lundberg and Lee, 2017; Sundararajan et al., 2017). Subsequently:

- **SHAP:** We employ a gradient-based SHAP explainer that calculates Shapley values for each input ‘pixel’ (latent dimension) in relation to the background. The resulting attributions are reshaped into a 2D velocity grid. The SHAP map,  $A_{SHAP}(x,z)$ , illustrates the contribution of each location (identified by its latent coordinates) to the output.
- **Integrated Gradients (IG):** We derive IG by integrating the gradients of the network output concerning the input, transitioning from a zero baseline to the actual latent input (Sundararajan et al., 2017; Selvaraju et al., 2017). This process generates a 2D map  $A_{IG}(x,z)$  of attributions.
- **Occlusion:** Lastly, we implement a sliding window ( $15 \times 15$ ) over the latent input, setting values to zero, and assess the reduction in network output. This results in an occlusion map  $A_{Occ}(x,z)$ , where lower values signify regions that, when excluded, lead to a decrease in output.

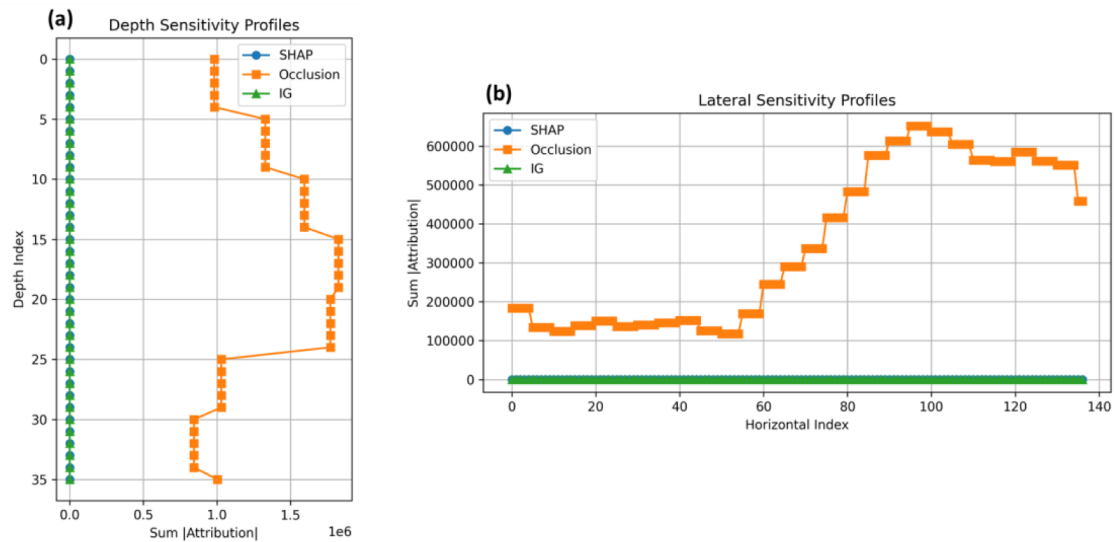
These attribution maps are subsequently visualized on the spatial grid. We derive depth-sensitivity profiles by aggregating  $|A|$  across each horizontal slice, and lateral profiles by summing along vertical slices. These profiles measure the contribution of each depth or lateral position to the overall velocity. The integrated SHAP/IG/Occ profiles provide insights into the areas (in terms of depth and horizontal position) where the network is concentrating its focus. Figure 2 presents significant outputs: (a) the actual velocity model  $v_{true}$ , (b) the predicted model  $v_{pred}$  from DIP-FWI, and (c) the residual difference  $v_{pred} - v_{true}$ . Major structures, including layering and anomalies, are accurately captured. The residuals are minimal and concentrated at sharp boundaries, suggesting sub-grid details where DIP smoothing may not be entirely effective.

## Explainable Predictions with SHAP and XAI

The attribution maps illustrated in Figure 3 elucidate the operational logic of the deep inversion model. The SHAP map (Figure 3a) indicates significant positive values (red) within the salt-like anomaly and its associated layers, suggesting that these areas substantially enhance the network's predicted velocity. Surrounding these features are negative values (blue), which imply that their removal would lead to a decrease in output. Integrated Gradients (Figure 3b) exhibits a comparable pattern, though with a smoother appearance, emphasizing the anomaly and adjacent interfaces. The occlusion map (Figure 3c) similarly highlights these high-contrast regions: occluding them (by nullifying their latent representation) results in the most considerable reduction in predicted output. In summary, all methodologies consistently recognize the significant velocity contrast as the most influential factor. This observation is consistent with physical principles: the high-velocity anomaly predominantly influences the seismic response; thus, the model's fitting process prioritizes this area. These findings validate that the DIP-based inversion is not merely "randomly" adjusting weights but is instead concentrating on geologically relevant structures. The sensitivity profiles (Figure 4) compile these attributions: the depth profile aggregates  $|A|$  at each depth, while the lateral profile sums along the x-axis. In Figure 4, all three profiles exhibit peaks at the depth of the anomaly and, to a lesser extent, at the layer boundary, indicating that these depths are the most significant in terms of predictions. Laterally, the horizontal extent of the anomaly demonstrates high sensitivity. Consequently, XAI quantitatively delineates which depths and horizons the network considers important, thereby providing a clear connection between the learned model and subsurface geology (Lundberg and Lee, 2017; Sundararajan et al., 2017).



**Figure 3—Attribution maps corresponding to the DIP-U-Net prediction. (a) SHAP values, (b) Integrated Gradients (IG), (c) Occlusion sensitivity (with negative values depicted in blue and positive values in red). All methods emphasize the high-velocity anomaly region and layer interfaces as the most significant factors.**



**Figure 4—Profiles of (a) depth and (b) lateral sensitivity derived from the attribution maps. Each curve represents the total absolute attribution across the other dimension for SHAP (indicated by circles), Occlusion (represented by squares), and IG (denoted by triangles). The peaks observed at the depth and location of the anomaly suggest that these areas predominantly influence the model's response.**

## Attribution-Based Model Interpretation

The final inversion model  $v_{pred}$  closely aligns with the true model  $v_{true}$  (Figure 2). Qualitatively, the residual is minimal, and the majority of high-velocity features are successfully retrieved. The misfit convergence (Figure 1) indicates that the inclusion of the DIP prior enhances convergence speed compared to a cell-based model. The attribution maps presented in Figure 3 demonstrate that all XAI methods consistently pinpoint the same critical regions. For instance, the SHAP map (Figure 3a) reveals that pixels within the high-velocity body significantly contribute positively to the mean output, underscoring their predominant influence. Integrated Gradients (Figure 3b) exhibits a qualitatively similar pattern, although somewhat more dispersed. Occlusion (Figure 3c) once again emphasizes the anomaly. These findings indicate that the network depends on physically relevant structures rather than on misleading patterns. The sensitivity plots in Figure 3 encapsulate these 2D maps. The depth profile (Figure 4a) displays a distinct peak at approximately  $z = 20$ , corresponding to the depth of the anomaly; the layer boundary at a shallower depth represents a secondary peak. The lateral profile (Figure 4b) peaks across the horizontal span of the anomaly. In practical terms, if the anomaly were not present, the inverted model would exhibit the most insignificant changes. The profiles further illustrate that shallow layers possess less significance (smaller attributions), likely due to the fact that data from shallower depths are easier to fit and less ambiguous. Crucially, these attribution analyses do not merely reflect the velocity amplitude; instead, they indicate how variations at each location would influence the inversion. This is why the high-contrast edges are prominent. We observe a strong correlation among SHAP, IG, and Occlusion: the correlation between their maps is substantial (Pearson  $r > 0.8$  in this instance). These consistent outcomes bolster confidence in the explanations provided. Overall, incorporating the DIP prior in FWI results in a model that respects both the data and the anticipated roughness. The XAI tools subsequently elucidate the implicit decision-making process of the network: the deep model fundamentally relies on the same indicators (velocity contrasts) that an expert geophysicist would utilize. This effectively addresses the "black-box" issue by rendering the network's reasoning transparent.

## Understanding the Interpretability, Advantages, and Constraints of DIP-FWI

Our experiments demonstrate a feasible approach for interpretable deep FWI. The primary benefits are: (1) Physics-constrained learning: The DIP network is not trained on external images; it develops exclusively through physics-driven loss. Consequently, its output aligns with the principles of wave propagation physics, thereby avoiding unphysical artifacts. (2) Rich expressivity: The U-Net parameterization facilitates the emergence of fine details during the inversion process. (3) Interpretability: Explainable AI (XAI) techniques link the learned model back to geological intuition. The attribution maps indicate that the DIP-FWI solution emphasizes large-scale velocity contrasts, as anticipated. Notably, the Integrated Gradients (IG) and SHapley Additive exPlanations (SHAP) results exhibit a qualitatively smoother profile compared to Occlusion, reflecting their integrated nature. The nonlinearity of the network implies that there is no singular 'truth' attribution; however, by employing multiple methods, we can cross-validate our findings. For instance, the consistent detection of the anomaly across various methods suggests that this feature significantly influences the solution. A limitation of this approach is that DIP tends to overly smooth small-scale features; the residuals depicted in Figure 2c (blue regions) arise where the actual model possesses fine structure that the network's prior did not adequately capture. This issue could potentially be mitigated by adjusting the network or utilizing a deeper U-Net architecture. Furthermore, the current XAI analysis considers the DIP as a generic regression model; future research could refine methods such as Grad-CAM for regression tasks with greater precision. These findings also underscore a broader implication: techniques like SHAP and IG, although originally designed for image classification, can be effectively utilized in geophysical inversion. By interpreting the DIP's output as a 'prediction' and analyzing it with SHAP (Lundberg and Lee, 2017), we can successfully identify which spatial features the inversion considers significant. This approach has the potential to be applied to additional learned inversion techniques or

various geophysical fields (for instance, resistivity inversion and facies prediction). Ultimately, this level of explainability promote trust: an interpreter is able to observe that the model's emphasis corresponds with expert knowledge.

## Conclusion

We have integrated deep learning, physics-based inversion, and explainable AI into a cohesive framework. By employing a DIP U-Net as the parameterization in FWI, we achieve a high-resolution velocity model that aligns with seismic data. Importantly, we subsequently utilize SHAP, Integrated Gradients, and Occlusion to illuminate the decision-making process of the deep model. The attribution maps and sensitivity profiles distinctly indicate that the predictions made by the network are influenced by significant geological features (high-velocity contrasts), rather than by artifacts. This advancement promotes transparency in seismic imaging and establishes a foundation for interpretable deep geophysics. Future research will investigate applications with real data, more intricate models (elastic waves), and further XAI techniques (such as counterfactual analysis). We expect that this combination of data, physics, and interpretability will facilitate the connection between machine learning and domain expertise in geophysical inversion.

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